

B.Stat.-401

Multivariate Analysis

Practical Problems

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Problem – 1: PCA

Simple PCA as an Example:

```
SBP <- c(126, 128, 128, 130, 130, 132)
DBP <- c(78, 80, 82, 82, 84, 86)
# Rename
x <- SBP; y <- DBP
plot(x, y, pch = 19)
S <- matrix(c(cov(x, x), cov(x, y), cov(y, x), cov(y, y)),
nrow=2, ncol=2, byrow=TRUE,
dimnames= list(c("x", "y"), c("x", "y")))
S
```

Problem - 1: PCA

```
a <- eigen(S)
a
# Centered the data
x1 <- x - mean(x)
x1
summary(x1)
y1 <- y - mean(y)
y1
summary(y1)
plot(x1, y1, pch = 19)
```

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Problem - 1: PCA

```
# The next step is to calculate the covariance matrix
cov(x1, x1)
cov(y1, y1)
cov(x1, y1)
m <- matrix(c(cov(x1, x1), cov(x1, y1), cov(y1, x1),
cov(y1, y1)),
nrow=2, ncol=2, byrow=TRUE,
dimnames= list(c("x", "y"), c("x", "y")))
m
```

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Problem - 1: PCA

```
# Next we need to find the eigenvector and eigenvalues
of the covariance matrix
e <- eigen(m)
e
# eigen() decomposition
# $values
# $vectors
#The largest eigenvalue is the first principal component; we
multiply the standardized values to the first eigenvector, which
is stored in e$vectors[,1].
pc1 <- x1 * e$vectors[1,1] + y1 * e$vectors[2,1]
Pc1
pc2 <- x1 * e$vectors[1,2] + y1 * e$vectors[2,2]
Pc2
```

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Problem - 1: PCA

```
Z <- data.frame(PC1 = pc1, PC2 = pc2)
plot(pc1, pc2, pch = 19)

cov.pc <- matrix(c(cov(pc1, pc1), cov(pc1, pc2),
  cov(pc2, pc1), cov(pc2, pc2)),
  nrow=2, ncol=2, byrow=TRUE,
  dimnames= list(c("pc1","pc2"), c("pc1","pc2")))
cov.pc
```

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Problem - 1: PCA

```
# Now to perform PCA using the prcomp() function.  
data <- data.frame(x,y)  
data.pca <- prcomp(data)  
data.pca  
names(data.pca)  
data.pca$x  
  
plot(data.pca$x[,1], data.pca$x[,2], pch = 19)  
# The sign is meaningless here. It's just chosen at  
# random at the beginning.  
# It does not change the relationship.
```

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Problem – 2: Factor analysis by SPSS

- Suppose an industry analyst would like to predict automobile sales from a set of predictors.
- However, many of the predictors are correlated, and the analyst fears that this might adversely affect her results.
- This information is contained in the sample data file, named [car_sales.csv](#).
- 157 observations with 10 variables (type, price, engine_s, horsepower, wheelbas, width, length, curb_wgt, fuel_cap, mpg)
- Use Factor Analysis to focus the analysis on a manageable subset of the predictors.

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Problem – 2: Factor analysis by SPSS

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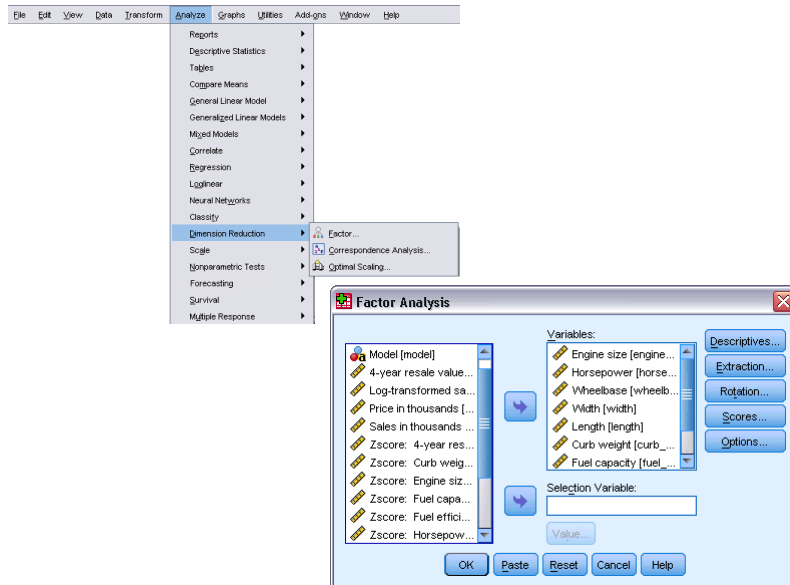
Problem – 2: Factor analysis by SPSS

- Data file is available at:
C > Program Files > IBM > SPSS > Statistics
> 20 > Samples > English > [car_sales.sav](#)
- Open the data file and do the following:
Analyze > Dimension Reduction > Factor...
- Select **Vehicle type** through **Fuel efficiency** as analysis variables

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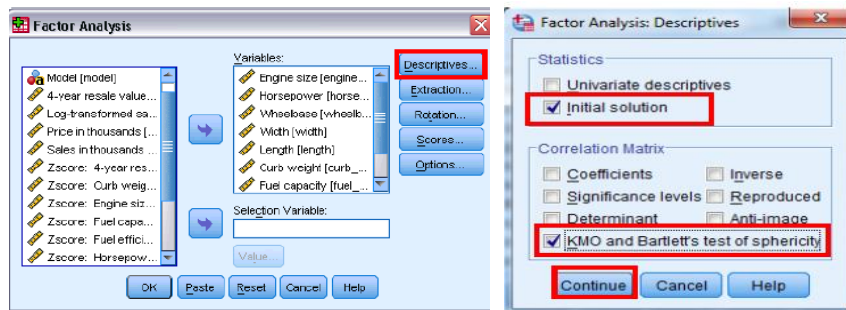
Problem – 2: Factor analysis by SPSS



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Problem – 2: Factor analysis by SPSS



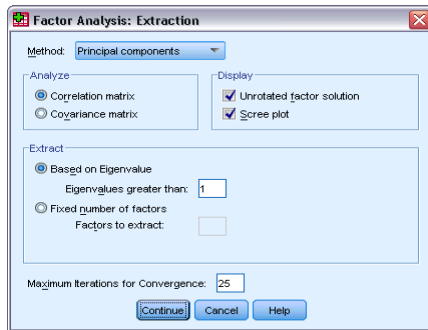
- This steps help to say about the adequacy of FA.
- Click **OK**

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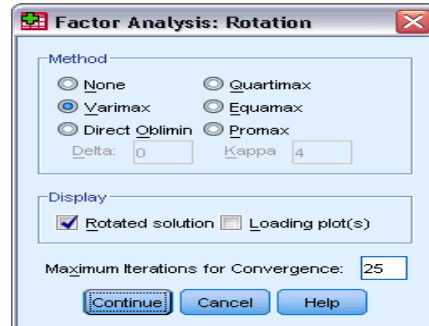
Problem – 2: Factor analysis by SPSS

- Click **Extraction**
- Select **Scree plot**
- Click **Continue**



This step helps to extract the number of factors.

- Again select **Rotation** in the Factor Analysis dialog box.
- Select **Varimax** in Method group.
- Click **Continue**.



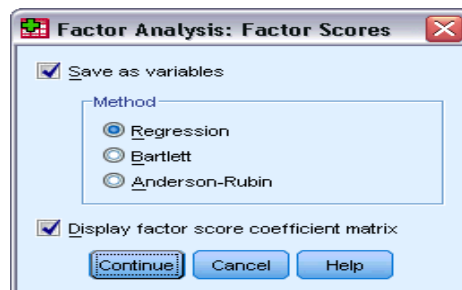
This step helps to rotate the factor loading (makes the loading very high or low) for clearly clustering the variables for each factor.

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Problem – 2: Factor analysis by SPSS

- Click **Scores** in the Factor Analysis dialog box.
- Select **Save as variables** and **Display factor score coefficient matrix**.
- Click **Continue**.
- Click **OK** in the Factor Analysis dialog box.



This steps compute the factor scores and put into the data file.

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Problem – 2: Factor analysis by SPSS

KMO and Bartlett's Test		
Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.833
Bartlett's Test of Sphericity	Approx. Chi-Square	1578.819
	df	45
	Sig.	.000

Kaiser-Meyer-Olkin (KMO) test: It *tests whether the data set is suitable* (or sufficiently correlated) *for factor analysis*. The value of KMO ranges between 0 to 1.

- High value of $KMO > 0.50$, generally indicate that a *factor analysis may be useful for the data set*.

Bartlett's test: It test the null hypothesis (H_0) that the *factors are insufficient to explain the variables under study*.

- Here the null hypothesis (H_0) is *rejected* by any level of significance ($p=0.000$).

Problem – 2: Factor analysis by SPSS

Communalities

	Initial	Extraction
Vehicle type	1.000	.930
Price in thousands	1.000	.876
Engine size	1.000	.843
Horsepower	1.000	.933
Wheelbase	1.000	.881
Width	1.000	.776
Length	1.000	.919
Curb weight	1.000	.891
Fuel capacity	1.000	.861
Fuel efficiency	1.000	.860

Communalities indicate the amount of variance in each variable that is explained by the factors.

➤ **Initial:** Initial communalities are estimates of the variance in each variable accounted for by all factors. *For principal components extraction, this is always equal to 1.0 for correlation analyses.*

➤ **Extraction:** Extraction communalities are estimates of the variance in each variable that explained by the selected 3 factors.

- The communalities in this table (for extraction) are all high, which indicates the extracted components represent the variables well.
- If any communalities are very low in a principal components extraction, you may need to extract another (one/more) component.

Problem – 2: Factor analysis by SPSS

Total Variance Explained

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	5.994	59.938	59.938	5.994	59.938	59.938	3.220	32.199	32.199
2	1.654	16.545	76.482	1.654	16.545	76.482	3.134	31.344	63.543
3	1.123	11.227	87.709	1.123	11.227	87.709	2.417	24.166	87.709
4	.339	3.389	91.098						
5	.254	2.541	93.640						
6	.199	1.994	95.633						
7	.155	1.547	97.181						
8	.130	1.299	98.480						
9	.091	.905	99.385						
10	.061	.615	100.000						

Extraction Method: Principal Component Analysis.

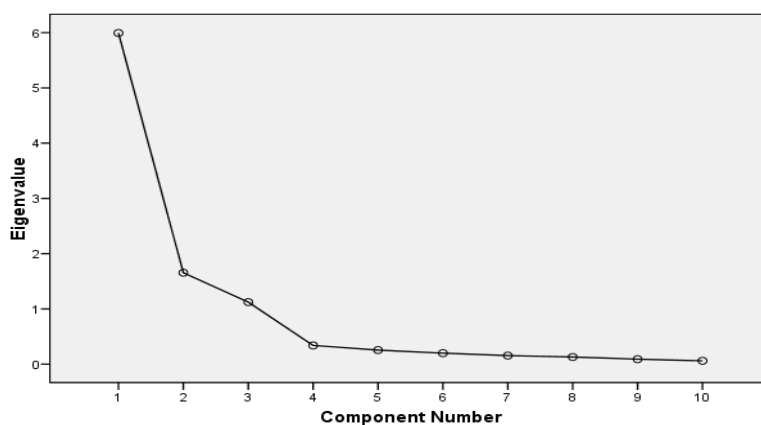
- The **Total** column gives the eigenvalue, or amount of variance in the original variables explained by each factor.
- The **% of Variance** column gives the percentage of the total variance of the original variables explained by each factor.
- The **Cumulative %** column gives the cumulative percentage of total variance accounted for by the factors.

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Problem – 2: Factor analysis by SPSS

Scree Plot



The last big drop occurs between the third and fourth components, so using the first three components is an easy choice.

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Problem – 2: Factor analysis by SPSS

Component Matrix^a

	Component		
	1	2	3
Vehicle type	.471	.533	-.651
Price in thousands	.580	-.729	-.092
Engine size	.871	-.290	.018
Horsepower	.740	-.618	.058
Wheelbase	.732	.480	.340
Width	.821	.114	.298
Length	.719	.304	.556
Curb weight	.934	.063	-.121
Fuel capacity	.885	.184	-.210
Fuel efficiency	-.863	.004	.339

Extraction Method: Principal Component Analysis.

a. 3 components extracted.

Rotated Component Matrix^a

	Component		
	1	2	3
Vehicle type	-.101	.095	.954
Price in thousands	.935	-.003	.041
Engine size	.753	.436	.292
Horsepower	.933	.242	.056
Wheelbase	.036	.884	.314
Width	.384	.759	.231
Length	.155	.943	.069
Curb weight	.519	.533	.581
Fuel capacity	.398	.495	.676
Fuel efficiency	-.543	-.318	-.681

Extraction Method: Principal Component Analysis.

Rotation Method: Varimax with Kaiser Normalization.

a. Rotation converged in 4 iterations.

- The first factor is highly correlated with **Price in thousands**, **Horsepower**, and **Engine size**
- The second factor is highly correlated with **Length**, **Wheelbase** and **Width**.
- The third factor is highly correlated with **Vehicle type**, **Fuel efficiency**, **Fuel capacity** and **Curb weight**.

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Problem – 2: Factor analysis by SPSS

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Extraction Method: Principal Component Analysis.

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- Let's give a possible interpretation to the factors. We might consider **factor 1 to be a "Price- Engine size and power" factor**.
- The **second factor** might be considered to be a **"Size of car"** factor.
- The **third factor** might be considered to be a **"Vehicle type- Fuel efficiency and capacity"** factor.

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Problem – 2: Factor analysis by SPSS

- One can reduce the size of the data file from ten variables to three components by using Factor Analysis with a principal components extraction.
- Note that the interpretation of further analyses is dependent upon the relationships defined in the rotated component matrix.
- This step of "translation" complicates things slightly, but the benefits of reducing the data file and using uncorrelated predictors outweigh this cost.

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Problem – 3: Factor analysis by R

- The previous automobile sales data will be used here.
- This information is contained in the sample data file, named [car_sales.csv](#).
- 157 observations with 10 variables (type, price, engine_s, horsepower, wheelbase, width, length, curb_weight, fuel_capacity, mpg)
- Use Factor Analysis to focus the analysis on a manageable subset of the predictors.

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Problem – 3: Factor analysis by R

Step-by-step Procedure of Exploratory Factor Analysis (EFA) in R:

- Step-1: Data Preparation
- Step-2: Checking Factorability (KMO & Bartlett's Test)
- Step-3: Determining the Number of Factors (Scree Plot, % variance explained)
- Step-4: Performing EFA (Using psych package)
- Step-5: Interpreting Results

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Problem – 3: Factor analysis by R

```
library(psych) # For EFA and factor analysis tools
car.data <- read.csv("car_sales.csv", as.is=TRUE)
head(car.data)
```

```
# Pearsons Correlation matrix
corr_matrix <- cor(car.data)
```

```
# Eigenvalues
ev <- eigen(corr_matrix, symmetric = TRUE)
e.value <- ev$values
```

```
# cumulative percentage of total variance
cum.e.value <- cumsum(e.value)
cum.e.value/sum(e.value)*100
```

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Problem – 3: Factor analysis by R

- Kaiser-Meyer-Olkin (KMO) Test of Sampling Adequacy. Measures sampling adequacy (values > 0.6 are acceptable).

`KMO(corr_matrix)`

- The overall KMO for our data is greater than 0.80, which is meritorious - this suggests that we can go ahead with our planned factor analysis.

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Problem – 3: Factor analysis by R

- Bartlett's Test of Sphericity. Test if the correlation matrix is an identity matrix (we want $p < 0.05$).

`cortest.bartlett(cor(car.data), n = nrow(car.data))`

- Scree plot

`plot(x = seq(1:length(e.value)), y = e.value, type = "o",
main="Scree Plot", xlab = "Number of factors", ylab = "Eigenvalue")`

- Number of factors based on total variation in eigenvalues.

`Nfacs <- 3`

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Problem – 3: Factor analysis by R

- Exploratory Factor Analysis (EFA) with 3 factors using the observed data

```
efa.3 <- factanal(x = car.data, factors = Nfacs,  
rotation="varimax")
```

```
efa.3
```

```
loadings <- efa.3$loadings # Factor loadings  
loadings
```

```
print(efa.3, digits=3, sort=TRUE)
```

- Interpretation – the same as Problem # 2.

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Problem – 4: ICA-01

Extracting the two independent tones from their linear mixture (generated additive Gaussian noise corrupted tones).

- a) Generate two Tones, s1 and s2, corrupted by the additive Gaussian noise, as follows:

```
s1 = 0.7*sin((1:1000)/19+0.57*pi) +mvrnorm(n=1000, mu=0,  
Sigma=0.004) # Tone 1
```

```
s2 = sin((1:1000)/33) + mvrnorm(n = 1000, mu = 0.03, Sigma  
= 0.005) # Tone 2
```

And assume a deterministic mixing matrix

$$A = \begin{bmatrix} 1 & 1.73 \\ -2 & 3.41 \end{bmatrix}$$

- b) Create Source matrix (S) and Mixing matrix (A) in R.

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Problem – 4: ICA - 01

- c) Mix two (signals) tones with the Mixing matrix (A) to get linearly mixed data (X).
- d) Draw scatter plots (joint distributions) of
 - (i) Source data with two independent components (S), and
 - (ii) Linearly mixed data (X).
- e) Estimate source tones by using fastICA algorithm based on X.
- f) Plot the observed and estimated tones and comment on the plot.

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Problem – 4: ICA - 01

Solutions:

```
a).
library(MASS) # To use mvrnorm()
set.seed(123)

# Tone 1 corrupted by noise
s1 <- 0.7*sin((1:1000)/19+0.57*pi) + mvrnorm(n =1000, mu=0,
Sigma= 0.004)
s1 <- as.numeric(s1)

# Tone 2 corrupted by noise
s2 <- sin((1:1000)/33) + mvrnorm(n = 1000, mu = 0.03, Sigma
= 0.005)
s2 <- as.numeric(s2)
```

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Problem – 4: ICA - 01

```
# Plot to see Original (source) Tones
par(mfcol = c(2, 1))
plot(s1, main="Additive Noise Corrupted Tone 1",
     xlab="Time", ylab = "Amplitude")
plot(s2, main = "Additive Noise Corrupted Tone 2", xlab =
     "Time", ylab = "Amplitude")
```

b).

```
S <- matrix(c(s1, s2), 1000, 2)
head(S,5)
```

c).

```
A <- matrix(c(1, 1.73, -2, 3.41), 2, 2, byrow = TRUE)
X <- S %*% A # Mixing Process:  $X = SA$ 
head(X)
```

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Problem – 4: ICA - 01

d).

```
dev.off() # reset the mfrow/mfcol parameter
# Scatter plot of S
plot(S, main = "Joint distribution of the source data with 2
independent components", xlab = "1st Dimension in S", ylab =
"2nd Dimension in S")

# Scatter plot of X
plot(X, main = "Joint distribution of the observed linearly
mixtures data, x1 and x2", xlab = "1st Dimension in X", ylab =
"2nd Dimension in X")
```

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Problem – 4: ICA - 01

e).

```
library(fastICA)
```

```
?fastICA
```

```
est <- fastICA(X, 2, alg.typ = "parallel", fun = "logcosh", alpha = 1,  
              method = "C", row.norm = FALSE, maxit = 200, tol = 0.0001,  
              verbose = TRUE)
```

```
str(est)
```

```
# Estimates of source tones
```

```
est$$
```

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Problem – 4: ICA - 01

f).

```
# Original and Estimated tones, S and  $\hat{S}$ :
```

```
dev.off() # reset the mfrow/mfcol parameter
```

```
ymin <- min(s1, s2, est$$[,1], est$$[,2])
```

```
ymax <- max(s1, s2, est$$[,1], est$$[,2])
```

```
plot(s1, col = "black", main = "Original and Estimated Tones",  
     xlab = "Time", ylab = "Amplitude", ylim = c(ymin, ymax))
```

```
lines(est$$[,1], col = "red") # first column of  $\hat{S}$ 
```

```
lines(s2, col = "green")
```

```
lines(est$$[,2], col = "blue") # second column of  $\hat{S}$ 
```

```
mtext("Original Tones in Black and Green, Estimated Tones in  
Red and Blue")
```

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Problem – 4: ICA - 01

The combination of 6 graphs - Original, Mixtures, and Estimated Tones:

```
par(mfcol = c(2, 3))
plot(1:1000, S[,1], type = "l", xlab = "Original, S1", ylab = "")
plot(1:1000, S[,2], type = "l", xlab = "Original, S2", ylab = "")
plot(1:1000, X[,1], type = "l", xlab = "Mixture, X1", ylab = "")
plot(1:1000, X[,2], type = "l", xlab = "Mixture, X2", ylab = "")
plot(1:1000, est$S[,1], type="l", xlab="Estimated, S1", ylab="")
plot(1:1000, est$S[,2], type="l", xlab="Estimated, S2", ylab="")
mtext("Comparison of Original, Mixture, and Estimated Tones",
side=3, line= -3, outer=TRUE)
```

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Problem – 4: ICA - 01

- As can be seen from this Figure, the estimated tones are very close to the original source tones (their signs are reversed, but this has no significance.)

Addition:

- # Linear mixing using random matrix
- A <- matrix(rnorm(4, mean = 1, sd = 10), nrow = 2, ncol = 2) # random matrix
- Repeat (a) to (f) using this random mixing matrix.

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Problem – 5: ICA - 02

An example of applying ICA to work with real audio files:

- Suppose two audio files, named `artificially_mixed1.wav` and `artificially_mixed2.wav` are given. Create mixed signals (linear combination of sources) with a matrix of ones or all-ones matrix as the mixing matrix. Apply ICA using the fastICA algorithm to separate the mixed sound.

```
# Load packages
library(fastICA)
library(tuneR)
library(seewave)
library(audio)
```

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Problem – 5: ICA - 02

```
# Load two or more source audio files
signal1 <- readWave("artificially_mixed1.wav")
signal2 <- readWave("artificially_mixed2.wav")

# Save sampling rates
samp_rate1 <- signal1@samp.rate
samp_rate2 <- signal2@samp.rate

# Plot the waves
plot(signal1)
plot(signal2)

# Keep just the actual signal
signal1 <- signal1@left
signal2 <- signal2@left
```

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Problem – 5: ICA - 02

```
# Check which one is shorter
length_of_shorter <- min(length(signal1), length(signal2))

# Combine the audio files into one matrix
signal_comb <- cbind(signal1[1 : length_of_shorter], signal2[1 :
length_of_shorter])
play(signal_comb)

# Use fastICA function to separate the signals into the 2 audio
files
separated <- fastICA(signal_comb, n.comp = 2, row.norm =
FALSE)
```

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Problem – 5: ICA - 02

```
# Extract new sounds
separated_1 <- separated$$[, 1] / max(separated$$[, 1])
separated_2 <- separated$$[, 2] / max(separated$$[, 2])

# Play sounds
play(separated_1)
play(separated_2)

# Save the new files
savewav(separated_1, f = samp_rate1, file = "separated_example1.wav")
savewav(separated_2, f = samp_rate2, file = "separated_example2.wav")
```

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Prob. 6: Canonical Correlation Analysis (CCA)

- A researcher has collected data on three psychological variables, four academic variables (standardized test scores) and gender for 600 college freshmen.
- She is interested in **how the set of psychological variables relates to the academic variables**.
- In particular, the researcher is interested in how many dimensions (canonical variables) are necessary to understand the association between the two sets of variables.

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Prob. 6: Canonical Correlation Analysis (CCA)

Description of the data

- We are going to investigate the associations between psychological measures and academic achievement measures.
- The data are given in the file, pmaam.csv, with 600 observations on eight variables.
- The **psychological variables** are:
 1. Control (means locus of control),
 2. Concept (means self concept) and
 3. Motivation (means self motivation).

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Prob. 6: Canonical Correlation Analysis (CCA)

- The **academic variables** are:
 1. Read (standardized tests in reading)
 2. Write (tests in writing)
 3. Math (tests in math) and
 4. Science (tests in science).
- Additionally, the variable
 1. Gender (is a zero-one indicator variable with the one indicating a female student).

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Prob. 6: Canonical Correlation Analysis (CCA)

- a) Read the data from the csv file and show the summary statistics of all variables.**

```
pmaam.data <- read.csv("pmaam.csv", as.is=TRUE)
head(pmaam.data)
summary(pmaam.data)
```

- b) Prepare Data for CCA: Split into two sets of variables, X for psychological variables and Y for academic performance variables.**

```
X <- pmaam.data[, 1:3] # psychological variables
Y <- pmaam.data[, 4:7] # academic variables
```

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c) Check correlations within sets and comment.

```
cat("Correlations within psychological variables:\n")  
print(round(cor(X), 3))
```

```
cat("\nCorrelations within academic variables:\n")  
print(round(cor(Y), 3))
```

```
cat("\nCorrelations between sets:\n")  
print(round(cor(X, Y), 3))
```

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d) Perform Canonical Correlation Analysis and display the results.

```
CCA.result <- cc(X, Y)  
print(CCA.result)
```

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e) Extract canonical correlations. Show the results of the test of significance of canonical correlations and comment.

```
library(CCP)
rho <- CCA.result$cor
n <- dim(pmaam.data)[1]
p <- ncol(X) # number of variables in first set
q <- ncol(Y) # number of variables in second set
```

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Calculate p-values using the F-approximations of different test statistics:

```
p.asym(rho, n, p, q, tstat = "Wilks")
p.asym(rho, n, p, q, tstat = "Hotelling")
p.asym(rho, n, p, q, tstat = "Pillai")
p.asym(rho, n, p, q, tstat = "Roy")
```

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Comment on the tests:

- The first test of the canonical dimensions tests whether all three dimensions are significant ($F = 12.77$, $p=0.000$).
- The next test tests whether dimensions 2 and 3 combined are significant ($F = 2.42$, $p=0.024$).
- Finally, the last test tests whether dimension 3, by itself, is significant ($F=0.15$, $p=0.860$), it is not.
- Therefore dimensions 1 and 2 must each be significant while dimension three is not.

Prob. 6: Canonical Correlation Analysis (CCA)

- In general, the number of canonical dimensions is equal to the number of variables in the smaller set; however, the number of significant dimensions may be even smaller.
- Canonical dimensions, also known as canonical variates, are latent variables that are analogous to factors obtained in factor analysis.
- For this particular model there are three canonical dimensions of which only the first two are statistically significant at the 0.05 level. .

Prob. 6: Canonical Correlation Analysis (CCA)

f) Compute and examine the canonical coefficients and canonical loadings.

The raw canonical coefficients:

```
CCA.result$xcoef
```

```
CCA.result$ycoef
```

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- The raw canonical coefficients are interpreted in a manner analogous to interpreting regression coefficients.
- That is, for the variable Read, a one unit increase in reading leads to a 0.044 decrease in the first canonical variate of set 2 when all of the other variables are held constant.
- Similarly, for the variable Science, a one unit increase in science leads to a 0.003 increase in the first canonical variate of set 2 when all of the other variables are held constant.

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Canonical loadings:

```
Loadings.X <- cor(X, CCA.result$scores$xscores)
Loadings.Y <- cor(Y, CCA.result$scores$yscores)

cat("Canonical loadings for psychological variables(X):\n")
print(round(loadings.X, 3))
cat("\nCanonical loadings for academic performance (Y):\n")
print(round(loadings.Y, 3))
```

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Cross-loadings:

```
Cross.loadings.X <- cor(X, CCA.result$scores$yscores)
Cross.loadings.Y <- cor(Y, CCA.result$scores$xscores)
cat("\nCross Loadings - X variables with Y canonical
variables:\n")
print(round(Cross.loadings.X, 3))
cat("\nCross Loadings - Y variables with X canonical
variables:\n")
print(round(Cross.loadings.Y, 3))
```

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- These loadings are correlations between variables and the canonical variates.
- The above correlations are between observed variables and canonical variables which are known as the canonical loadings.
- These canonical variates are actually a type of latent variable.

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- g) Visualization: (i) Plot first pair of canonical variables,
(ii) Plot second pair of canonical variables.**

(i) Plot of first pair of canonical variables

```
plot(CCA.result$scores$xscores[, 1],  
     CCA.result$scores$yscores[, 1],  
      xlab = "First psychological canonical variable",  
      ylab = "First academic canonical variable",  
      main = "First canonical pair",  
      pch = 19, col = "blue", alpha = 0.6)  
abline(lm(CCA.result$scores$yscores[, 1] ~  
          CCA.result$scores$xscores[, 1]),  
       col = "red", lwd = 2)
```

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(ii) Plot of second pair of canonical variables

```
plot(CCA.result$scores$xscores[, 2],  
     CCA.result$scores$yscores[, 2],  
      xlab = "2nd psychological canonical variable",  
      ylab = "2nd academic canonical variable",  
      main = "Second canonical pair",  
      pch = 19, col = "blue", alpha = 0.6)  
abline(lm(CCA.result$scores$yscores[, 2] ~  
          CCA.result$scores$xscores[, 2]),  
       col = "red", lwd = 2)
```

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--- More ---

--- Incomplete ---

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Thank you

Any Questions?